

$[[0, 1], [2, 3], [4, 5]]$

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$[[0, 1, 2], [3, 4, 5]]$

$x \text{ rows } y \text{ cols} \Rightarrow x \cdot y = n$

$[[[0], [1], [2]], [[3], [4], [5]]]$

$1 \mid n$ For any integer n
 $\forall n \in \mathbb{Z}$

Prove: $x = y \bmod n$ and $y = z \bmod n \Rightarrow x = z \bmod n$

Assume $x = y \bmod n$ and $y = z \bmod n$

Want to show $x = z \bmod n$

By def $\exists k_1$ s.t. $x - y = k_1 n$

$\exists k_2$ s.t. $y - z = k_2 n$

Want to show $\exists k$ s.t. $x - z = k n$

[Aside] $27 = 47 \bmod 10$ $47 = 57 \bmod 10$

$$-20 = 27 - 47 = -2 \cdot 10 \quad -10 = 47 - 57 = -1 \cdot 10$$

$$-30 = 27 - 57 = -3 \cdot 10 = (-2 + -1) \cdot 10$$

Algebra $(x - y) + (y - z) = x - z$

$$k_1 n + k_2 n = x - z$$

$$(k_1 + k_2) n = x - z$$

Let $k = k_1 + k_2$



Prove $x = x \pmod n$

$$\exists k \text{ s.t. } x - x = kn$$

(there exists) $\uparrow$ "Such that" $0 = kn$ End proofs
 $\downarrow$

Let $k=0$ then $0 = kn$. Q.E.D

Prove $x = y \pmod n \Rightarrow y = x \pmod n$

Assume $x = y \pmod n$. Want to show $y = x \pmod n$
By def, $\exists k$ s.t. $x - y = kn$

Want to show $\exists k_1$ s.t. $y - x = k_1 n$

$\uparrow$ introduce notation to avoid conflicts

Let $k_1 = -k$ then

$$x - y = kn \Rightarrow y - x = -kn$$

$$y - x = (-1)(x - y) \Rightarrow y - x = k_1 n$$

$$= (-1) kn$$

Q.E.D

$$= -k n$$

$$27 = 47 \pmod{10}$$

$$27 - 47 = -2(10) \text{ Use words}$$

$$47 - 27 = 2(10)$$

$$47 = 27 \pmod{10}$$

